

A Conductive Self-Healing Hybrid Gel Enabled by Metal–Ligand Supramolecule and Nanostructured Conductive Polymer

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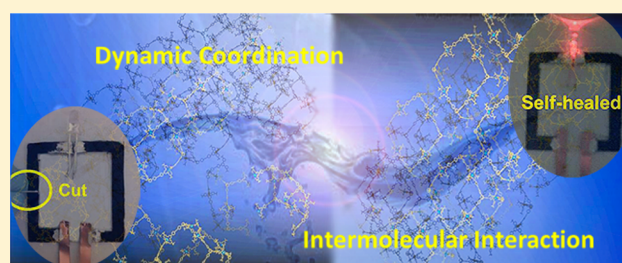
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S Supporting Information

ABSTRACT: Self-healing materials emerge as a fascinating class of materials important for various technological applications. However, achieving the synergistic characteristics of high conductivity, room-temperature self-healing ability, and decent mechanical properties still remains a critical challenge. Here we develop for the first time a hybrid gel based on self-assembled supramolecular gel and nanostructured polypyrrole that synergizes the dynamic assembly/disassembly nature of metal–ligand supramolecule and the conductive nanostructure of polypyrrole hydrogel and exhibits features of high conductivity (12 S m^{-1}), appealing mechanical and electrical self-healing property without any external stimuli, and enhanced mechanical strength and flexibility. The attractive characteristics of the hybrid gel are further demonstrated by a flexible yet self-healable electrical circuit. Our work shows the great potential of self-healing hybrid gel system in flexible electronics and provides a useful strategy to design multifunctional self-healing materials.

KEYWORDS: Supramolecular chemistry, hybrid gel, self-healing, nanostructured conductive polymer, multifunctional



Self-healing materials with conductive properties have attracted growing interest in both academia and industry due to their potential applications in a broad range of technologies, such as self-healing electronics,¹ medical devices,² artificial skins,³ and soft robotics.⁴ For practical applications, these materials should demonstrate good conductivity and repeatable mechanical and electrical self-healing properties at room temperature, as well as decent mechanical strength and flexibility, to meet the requirements for fabrication of flexible devices.⁵

Great efforts have been dedicated to developing conductive self-healing materials. White, Moore, and co-workers⁶ developed the use of microcapsules containing liquid precursor healing agents for structural healing. In these systems, the local healing agent is depleted after capsule rupture. Bao et al.³ demonstrated an alternative approach by combining a supramolecular organic polymer and nickel microparticles, resulting in a composite with mechanical and electrical self-healing properties at ambient conditions, whereas a large number of inorganic particles are needed for the preparation of composite. Recently, Park et al.⁷ synthesized a conductive and self-healing hydrogel by polymerization of pyrrole within agarose matrix. The self-healing behavior of the resultant composite, however, can only be excited under external thermal or optical stimuli. Therefore, the development of self-healing, highly conductive, mechanically strong, and lightweight materials remains a critical challenge.⁸

In the past decades, the supramolecular chemistry has witnessed rapid development of metallo-supramolecular structures based on the highly directional and predictable feature of metal-mediated self-assembly.⁹ Driven by directional and conjugated structures and intermolecular forces, these supramolecular structures could further hierarchically self-assemble into higher order nanostructures, that is, supramolecular gels.¹⁰ More importantly, due to the moderate bond energy of metal–ligand bonds and noncovalent interactions among supramolecules, the supramolecular gels can dynamically assemble or disassemble, associate or dissociate at room temperature, thus showing features such as self-healing property and sol–gel phase transitions.¹¹ Recently, conductive polymer hydrogels (CPHs) such as polyaniline (PANI) and polypyrrole (PPy) hydrogels have been synthesized using phytic acid as the gelator and dopant.¹² The framework of the resulted CPHs provides ideal 3D interconnected paths for electron transport, thus reaching a conductivity as high as 11 S m^{-1} .¹² Such 3D hierarchically porous structures offer large open channels to support the introduction of second gel component and provide an ideal interface between conductive hydrogels and other synthetic systems.¹³ However, the fragile nature and lack of self-healing property inhibits CPHs' further applications.

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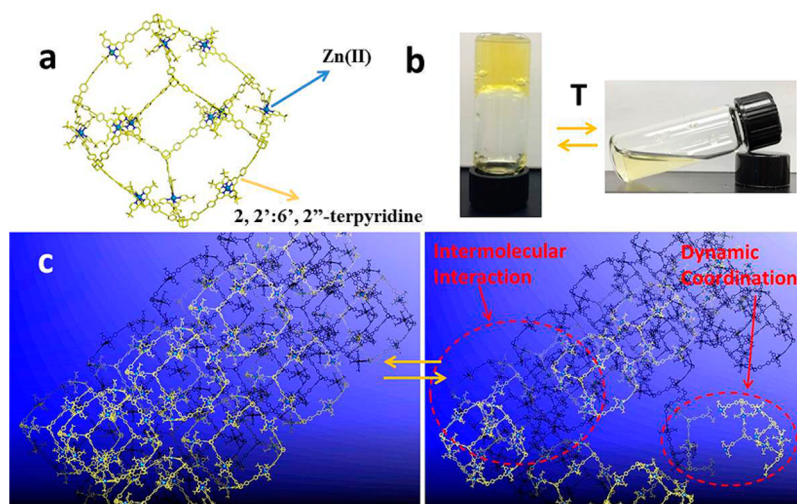


Figure 1. Supramolecular gel. (a) The molecular structure of Zn-tpy supramolecule. (b) G-Zn-tpy shows a reversible sol–gel phase transition at 50 °C above which G-Zn-tpy becomes a homogeneous solution. (c) Schematic illustration of proposed mechanisms of supramolecular gels' self-healing behavior. The dynamic intermolecular interaction and coordination at the crack position helps heal the gel material.

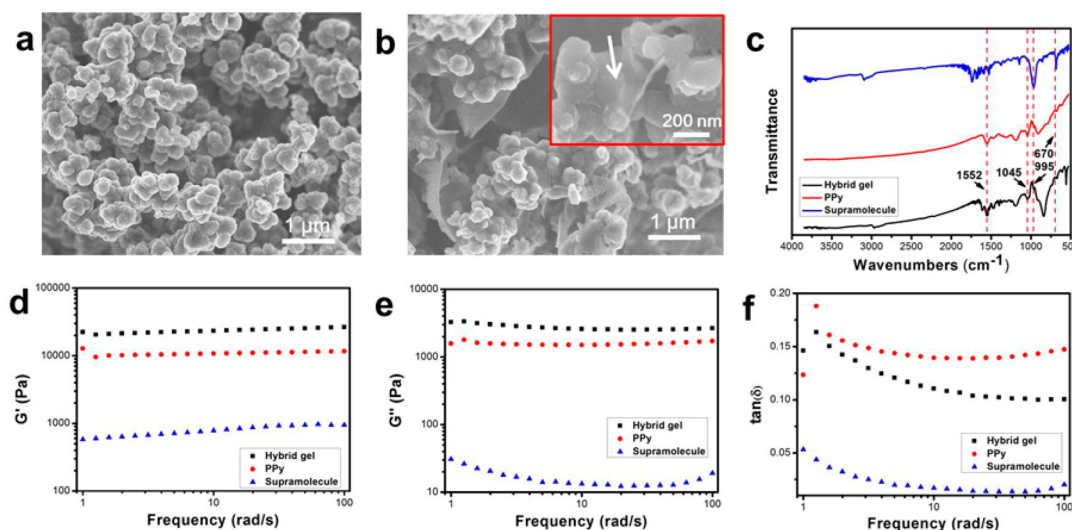


Figure 2. Characterization of PPy/G-Zn-tpy hybrid gel. (a,b) SEM images of PPy, and PPy/G-Zn-tpy hybrid gel, respectively. (c) FTIR spectra of Zn-tpy supramolecule, PPy, and PPy/G-Zn-tpy hybrid gel. (d,e) The storage modulus (G') and loss modulus (G'') of Zn-tpy supramolecule, PPy, and PPy/G-Zn-tpy hybrid gel (f) The tangent of the phase angle (G''/G') values of Zn-tpy supramolecule, PPy, and PPy/G-Zn-tpy hybrid gels.

Herein, we synthesized an acetonitrile-based supramolecular gel with cubic architecture, namely, G-Zn-tpy by using 2,2':6',2''-terpyridine (tpy) as organic ligand and Zn(II) as gluing element and developed a simple strategy in which the supramolecular gel was introduced into CPH matrix by taking advantage of its reversible sol–gel transition, resulting in a hybrid gel material of PPy/G-Zn-tpy. In this hybrid gel system, the nanostructured PPy gel constructs a 3D conductive network to promote the transport of electrons and mechanically reinforce the hybrid gel while the supramolecular gel contributes to self-healing property and also conductivity. More importantly, the design of hybrid gel also excites synergic effects such as enhanced mechanical strength and elasticity. With these features, films of hybrid gel on various substrates are fabricated and demonstrated to possess properties of room-temperature self-healing, high conductivity, and good flexibility, thus becoming a promising material for various applications such as self-healing electronics, artificial skins, soft robotics, and

biomimetic prostheses. The “guest-to-host” strategy by taking advantage of chemical and physical features of each component is also demonstrated to be an important and fundamental way to design and synthesize a class of functional polymeric materials.

The synthesis of Zn-tpy supramolecule was described in our previous paper.¹⁴ The supramolecule was assembled by 12 metal ions and 8 three-armed building blocks that were designed by linking three tpy groups to rigid adamantane core, resulting in a structure of cubic cage (Figure 1a). In the subsequent research, the gelation behavior of Zn-tpy supramolecule was discovered with a critical gelator concentration of 5 wt % (in acetonitrile, room temperature). The formation of supramolecular gel is driven by the intermolecular interactions, such as hydrophobic forces and π – π stacking.^{10b} The cubic cage structure also enhances the hierarchical self-assembly to form gel, because with such rigid and uniform geometry the supramolecules could be well arranged along three-dimensional

directions^{10b,15} (the gelation behavior was not observed for the other two supramolecules with bipyramidal-like dimer and tetrahedron structures synthesized by us). In addition, the G-Zn-tpy shows a reversible sol–gel phase transition at 50 °C above which G-Zn-tpy becomes a homogeneous solution, as shown in Figure 1b. The gel to sol transition could be explained by the disassembly of supramolecules, as well as the dissociation of metal–ligand bonds at high temperature. More importantly, the assembly or disassembly, association or dissociation behaviors are dynamic processes due to relatively weak noncovalent intermolecular interactions and moderate bond energy (15–50 kcal mol^{−1}) of metal–ligand bonds.^{11b,c} The dynamic nature endows the supramolecular gel good self-healing property at room temperature. Figure 1c shows the schematic illustration of self-healing behavior of G-Zn-tpy in which the combination of dynamic intermolecular interaction and coordination drives the reassembly of supramolecules at the crack, thus self-healing small cracks within tens of seconds and big cracks in minutes.

By taking advantage of the sol–gel transition property of G-Zn-tpy, hybrid gel could be readily synthesized. In a typical synthesis, PPy hydrogel is first synthesized using phytic acid as the gelator and dopants.¹⁶ The obtained hydrogel is freeze-dried to form an aerogel, which maintains its high surface area and three-dimensional porous nanostructures (Supporting Information Figure S1). The PPy aerogel could swell in the supramolecule solution above 50 °C (the weight ratio of PPy to G-Zn-tpy is 6:1; the weight ratio of PPy in the hybrid gel is ~83%). As the swollen gel cools, the supramolecular gel in situ forms within the PPy matrix, resulting in a binary network gel. Figure 2a,b shows the morphologies and microstructures of PPy aerogel and PPy/G-Zn-tpy hybrid gel examined by scanning electron microscope (SEM), respectively. PPy aerogel is formed by interconnected spherulike particles with size ranging from 150 to 300 nm, resulting in a hierarchically porous structure in which two levels of pores could be observed for the conductive polymer hydrogel, including the pores between the nanospheres, and the bigger micron size pores between domains of nanospheres. After introducing the supramolecule gel in PPy matrix, thin film-like structures could be found (inset in Figure 2b shows a typical film structure of supramolecule gel). The supramolecular gel wraps the PPy particles and further connects the PPy domains together, building up a secondary self-cross-linked network to form a binary network gel. The chemical structure of as-prepared hybrid gel was examined by Fourier transform infrared spectroscopy (FTIR). Figure 2c shows the FTIR spectra of G-Zn-tpy, PPy, and PPy/G-Zn-tpy hybrid gel. The spectrum of PPy shows the absorption peak at 1552 cm^{−1} corresponding to the in-ring stretching of C=C bonds in the pyrrole rings and the peak at 1045 cm^{−1} that can be attributed to the in-plane deformation of N–H bond.^{16a} For the spectrum of G-Zn-tpy, two major characteristic peaks could be found at 995 and 670 cm^{−1} which can be attributed to the breathing mode of pyridine and vibration of Zn–N bonds, respectively.¹⁷ All the characteristic peaks of PPy and G-Zn-tpy can be found in the FTIR spectrum of synthesized hybrid gel of PPy and G-Zn-tpy, confirming the formation of composites. We can also notice that the characteristic peaks of G-Zn-tpy slightly shift toward the lower wavenumber side, indicating the interactions between Zn-tpy molecule and PPy backbone that could weaken the bonding strength of supramolecule due to the effect of delocalization of the π -electrons.¹⁸

Desirable mechanical properties, especially mechanical strength and elasticity, are important for gel materials to be applied in practical devices.¹⁹ Gels are viscoelastic materials and that exhibit the properties of storing and dissipation of energy.²⁰ The amount of energy stored in the gel system and the amount of energy dissipated within the system under the oscillatory stress are indicated by the storage modulus (G') and the loss modulus (G''), respectively. The G' and G'' values of G-Zn-tpy gel with acetonitrile as solvent, PPy aerogel, and PPy/G-Zn-tpy hybrid gel with acetonitrile as solvent are shown in Figure 2d,e. Their gel states are revealed by the wide linear viscoelastic region in the dynamic frequency sweep experiments and further confirmed by the fact that the value of storage modulus is higher than that of the loss modulus in each case. Although the pure G-Zn-tpy is rather weak according to its low G' and G'' values, the G' value of PPy/G-Zn-tpy hybrid gel is significantly enhanced and even higher than that of PPy, indicating the increased strength. This enhanced mechanical strength is attributed to the reinforcement effect brought by the continuous and mechanically efficient network of PPy with rigid backbones^{13,21} and the binary network structure of hybrid gel in which the self-cross-linked supramolecule gel warps and links the PPy particles together. The covalent bond of PPy chains and metal–ligand bond of supramolecule facilitate the dissipation of energy. The interactions between two networks of gels also contribute to the increase of mechanical strength of hybrid gels.¹³ The tangent of the phase angle—the ratio of viscous modulus (G'') to elastic modulus (G') is a useful quantifier of the presence and extent of elasticity in a gel system.²² The $\tan \delta$ values of less than unity indicate elastic-dominant behavior and values greater than unity indicate viscous-dominant behavior. Figure 2f shows that all three samples are elastic-dominant gel systems while the pure Zn-tpy gel is most elastic. Compared to the rigid and fragile PPy hydrogel, the PPy/G-Zn-tpy hybrid gel is much more elastic according to their decreased $\tan \delta$ value, thus possessing the ability to be applicable and processable in flexible devices. Simply dispersing PPy in supramolecular gel will result in products with low mechanical strength and conductivity (Supporting Information Figure S4).

With the improved mechanical properties, thin films of PPy/G-Zn-tpy hybrid gel were successfully fabricated on flexible substrates such as PDMS and Kapton films that are commonly used substrates for flexible electronic devices. The PPy aerogel was first polymerized on flexible substrates and then supramolecule gel was introduced to form the hybrid gel film. The thickness of films could vary from ~0.1 to 0.5 mm by controlling the amount of PPy precursor and all the films are uniform and with good quality. The high flexibility of PPy/G-Zn-tpy hybrid gel films were demonstrated by the fact that the films can maintain uniformity and adhere firmly on substrates when they were significantly bended. As the controlled samples, PPy aerogel films were also bended but resulted in the formation of cracks and peeling off from substrates (Supporting Information Figure S2). In addition to thin films on flexible substrates, free-standing films of PPy/G-Zn-tpy hybrid gel were also prepared and shaped to various morphologies, revealing good mechanical strength and processability of PPy/G-Zn-tpy hybrid gel. The properties of PPy/G-Zn-tpy hybrid gel films are consistent with the results of rheology tests, indicating its enhanced mechanical strength and elasticity.

The conductivity of PPy/G-Zn-tpy hybrid gel film was tested by using four-probe method and can reach as high as 12 S m^{−1}

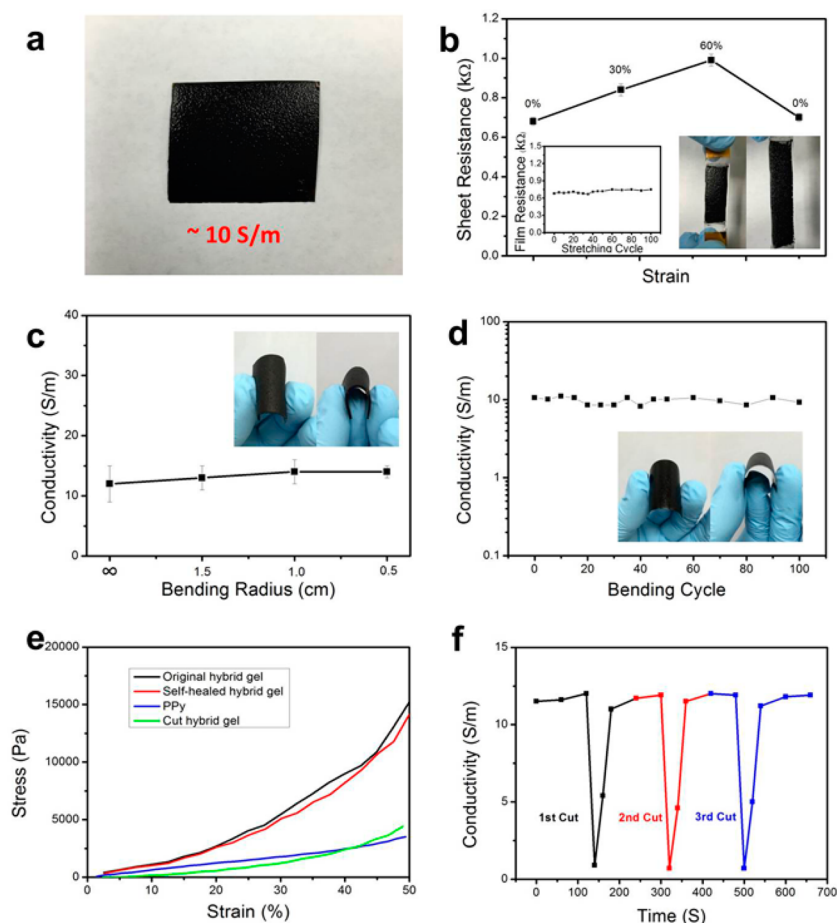


Figure 3. Thin film of PPy/G-Zn-tpy hybrid gel. (a) Photograph of PPy/G-Zn-tpy hybrid gel thin film coated on Kapton substrate. (b) Film resistance of PPy/G-Zn-tpy hybrid gel thin film coated on PDMS substrate under different stretching states. Inset (left) shows the film resistance after different stretching cycles and inset (right) shows the optical images of PPy/G-Zn-tpy hybrid gel thin film at initial status and 67% strain. (c) Conductivity of PPy/G-Zn-tpy hybrid gel thin film under different bending states. Inset shows the optical images of bended PPy/G-Zn-tpy hybrid gel thin film coated on PDMS substrate. (d) Conductivities of PPy/G-Zn-tpy hybrid gel thin film after different bending cycles. Inset shows the optical images of bended PPy/G-Zn-tpy hybrid gel thin film coated on Kapton substrate. (e) Compression test for different samples. (f) The conductivities of hybrid gel at different stages during cutting and self-healing processes. The cut samples were physically contacted to each other.

(Figure 3a) which is comparable to pure conductive polymers and among the highest values of conductive hybrid gels.¹³ The excellent electrical properties of our hybrid gels could be attributed to the interconnected networks of conductive polymers that can act as the continuous transporting path for electrons, as well as the conductive nature of G-Zn-tpy with conductivity of $\sim 5 \text{ S m}^{-1}$. The mechanical stability of electrical properties of PPy/G-Zn-tpy hybrid gel film was demonstrated by bending and stretching tests. Figure 3b shows the result of stretching test using PDMS as the substrate. The hybrid gel film was stretched to 33% and 67% strain and the film resistance increased from 0.68 to 0.84 and 0.99 k Ω , respectively. The increase was caused by the formation of small cracks in the film during stretching. However, when the film was released to its original length, the resistance decreased to 0.70 k Ω . Optical images of stretching test were shown in Supporting Information Figure S3. The recovery could be explained by the high elasticity and self-healing property of PPy/G-Zn-tpy hybrid gel. The PPy/G-Zn-tpy hybrid gel film shows good mechanical stability during the bending tests in which the conductivity of PPy/G-Zn-tpy hybrid gel film remains almost constant under different bending radii, that is, 1.5, 1.0, and 0.5 cm (Figure 3c). The conductivity of PPy/G-Zn-tpy hybrid gel film can also

maintain $\sim 10 \text{ S m}^{-1}$ up to 100 bending cycles as shown in Figure 3d. The electrical stability of PPy/G-Zn-tpy hybrid gel film was supported by its enhanced mechanical strength and elasticity that help maintain the integrity of 3D continuous network for electron transport during mechanical deformation and ensure the good contact between gel film and flexible substrates.

The self-healing property is another key feature of G-Zn-tpy. The PPy/G-Zn-tpy hybrid gel inherits this unique feature and is able to self-heal the cracks at room temperature. We conducted compression tests on the samples at different states (original status, right after cutting to pieces, self-healed status) and compared the compression strength and modulus (Figure 3e). The images of hybrid gel at different states were shown in Supporting Information Figure S5. It can be noted that because the cutting direction is parallel to the compression direction, the sample tended to be extruded out during the test, thus further breaking the sample. We can see that the original hybrid gel and self-healed gel show similar behavior in compression test while the cut sample shows much lower strength, especially at high strain. The modulus of original gel, self-healed gel, and cut gel were calculated to be 107, 101, and 29 Pa, respectively. As a reference, we also conducted the compression test for pure

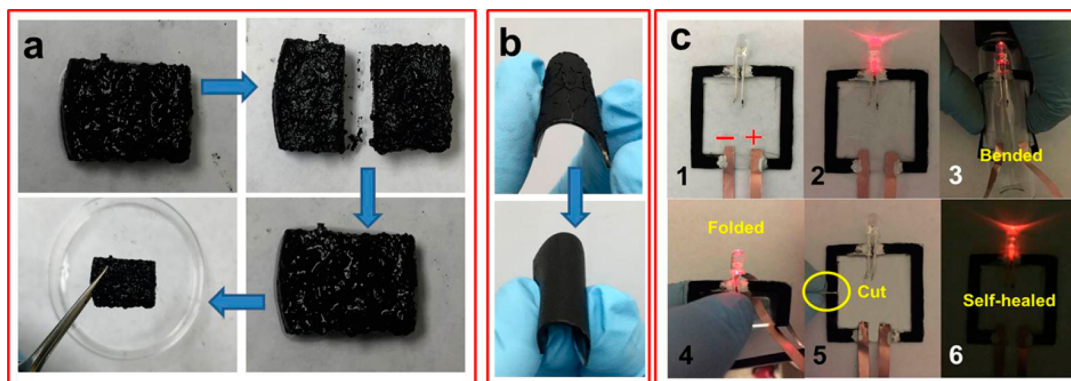


Figure 4. Self-healing property of PPy/G-Zn-tpy hybrid gel. The self-healing behavior of PPy/G-Zn-tpy hybrid gel: (a) bulk sample was cut into half and then placed together. After 1 min, the sample self-healed to be an integrated one and can support its own weight when lifted by a tweezers. (b) The initially cracked PPy aerogel film was self-healed after introducing G-Zn-tpy. (c) Self-healing circuit based on PPy/G-Zn-tpy hybrid gel: (c1,c2) optical images of circuit based on PPy/G-Zn-tpy hybrid gel film at open and closed states; (c3,c4) the circuit functions well under bended and folded states. (c5,c6) The self-healing behavior of designed circuit: the left side of PPy/G-Zn-tpy hybrid gel film was cut and the circuit became open and the bulb was extinguished. After 1 min of self-healing, the circuit was re-established and the LED bulb could be lighted again.

PPy gel and its modulus was 74 Pa. The results showed that after cutting to pieces, the gel was much weaker because it could not relieve the stress well along the horizontal direction. However, the self-healed gel was mechanically integrated and recovered to its initial strength. The self-healing property was also demonstrated by the measurement and comparison of conductivities of hybrid gel before and after cutting (Figure 3f). We measured the conductivity of the sample every 60 s between each cut and every 20 s during its self-healing process. We can see that after cutting, the conductivity of hybrid gel is much lower than that of original gel even though the two half samples were physically contacted to each other. After around 1 min, the conductivity recovered to initial value due to the reformation of an integrated conductive 3D network.

We further demonstrated the self-healing behavior of PPy/G-Zn-tpy hybrid gel for both bulk and thin film samples. Initially, one piece of PPy/G-Zn-tpy hybrid gel was shaped to a box with section area of $\sim 1 \text{ cm}^2$ and cut into half as shown in Figure 4a. Then two separated pieces were placed together and after 1 min, the sample self-healed to form an integrated one and can support its own weight when lifted by a pair of tweezers. More importantly, the sample was also electrically healed. The resistances of the samples before and after cutting remained the same. To show the self-healing behavior of PPy/G-Zn-tpy hybrid gel film (Figure 4b), we prepared the nanostructured PPy first and intentionally made some cracks by bending the PPy film. Then a solution of Zn-tpy supramolecule was introduced into this cracked thin film and after the formation of G-Zn-tpy, the cracks disappeared rapidly. The self-healed thin film of PPy/G-Zn-tpy hybrid gel exhibited excellent flexibility and high conductivity of $\sim 10 \text{ S m}^{-1}$. The self-healing property of PPy/G-Zn-tpy hybrid gel is brought by the supramolecule gel which acts as a dynamic “glue” in the hybrid system due to its dynamic assembly/disassembly, association/dissociation behaviors based on relatively weak noncovalent intermolecular interactions and moderate bond energy of metal–ligand bonds. According to the microstructure of PPy/G-Zn-tpy hybrid gel shown in SEM images, the self-cross-linked supramolecule gel helps to support the PPy network by wrapping and connecting PPy particles together, as well as building a self-supported network. The unique binary network structure is advantageous for the self-healing behavior of hybrid gel. The PPy matrix also plays an important role because its hierarchically porous

structure could facilitate the molecule motion and transportation at the cracks and its tough nature favors the mechanical healing behavior.²³

To further demonstrate the mechanical stability, high conductivity, and self-healing feature of PPy/G-Zn-tpy hybrid gel, we designed a complete circuit composed of a LED bulb with driving voltage of 3.0–3.2 V as the electrical load, PPy/G-Zn-tpy hybrid gel film as the conductor and two dry batteries (1.5 V) as the power source. As shown in Figure 4c, a circle of PPy/G-Zn-tpy hybrid gel thin film was coated on PDMS substrate, then the LED bulb was involved and the power source was linked into the circuit by two copper wires. The attaching points of LED bulb and copper wires were strengthened by coating silver paste to enhance the contacts. The bulb was successfully lighted when the circuit was switched to close status, indicating the high conductivity of our PPy/G-Zn-tpy hybrid gel because there is no apparent voltage drop on the hybrid gel part. Notably, this circuit could work well under bended, even folded states, showing the outstanding flexibility and mechanical strength of the PPy/G-Zn-tpy hybrid gel. At last, the self-healing property of PPy/G-Zn-tpy hybrid gel was also demonstrated on the circuit. As shown in Figure 4c5,c6, in a typical test the left side of PPy/G-Zn-tpy hybrid gel film was cut and the circuit then became open and the bulb was extinguished. After about 1 min, the circuit was re-established and the LED bulb could be lighted up again. This self-healing test was successfully conducted for multiple times at the same position and also at other positions. This working circuit shows the great potential of PPy/G-Zn-tpy hybrid gel for practical applications such as self-healing electronics, biosensors, and artificial skins.

In summary, we developed a promising self-healing PPy/G-Zn-tpy hybrid gel by incorporating a G-Zn-tpy supramolecule within the PPy aerogel matrix. The hybrid gel combines the high conductivity of PPy aerogel and self-healing property of G-Zn-tpy and exhibits enhanced mechanical strength and excellent elasticity due to its unique binary network structure. The self-healing behavior is efficient and could be observed at room temperature without any external stimuli, owing to dynamic assembly or disassembly of supramolecules and association or dissociation of metal–ligand bonds. Supported by these unique features, thin films with good uniformity and flexibility of G-Zn-tpy hybrid gel were successfully fabricated

and a self-healable electronic circuit based on hybrid gel films was demonstrated. This conductive, room-temperature self-healing gel material takes unique advantage of supramolecular chemistry and polymer nanoscience and shows its potential applications in various fields such as self-healing electronics, artificial skins, soft robotics, biomimetic prostheses, and energy storage.²⁴

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.5b03069.

Detailed experimental procedures and supplementary characterization methods including stretching and compression tests of hybrid hydrogel samples. (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) (a) Wang, H.; Zhu, B.; Jiang, W.; Yang, Y.; Leow, W. R.; Wang, H.; Chen, X. *Adv. Mater.* **2014**, *26*, 3638–3643. (b) Wang, C.; Wu, H.; Chen, Z.; McDowell, M. T.; Cui, Y.; Bao, Z. *Nat. Chem.* **2013**, *5*, 1042–1048.
- (2) (a) Zhang, Y.; Yang, B.; Zhang, X.; Xu, L.; Tao, L.; Li, S.; Wei, Y. *Chem. Commun.* **2012**, *48*, 9305–9307. (b) Holten-Andersen, N.; Harrington, M. J.; Birkedal, H.; Lee, B. P.; Messersmith, P. B.; Lee, K. Y. C.; Waite, J. H. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108*, 2651–2655. (c) Irwansyah, I.; Li, Y.-Q.; Shi, W.; Qi, D.; Leow, W. R.; Tang, M. B. Y.; Li, S.; Chen, X. *Adv. Mater.* **2015**, *27*, 648–654.
- (3) Tee, B. C. K.; Wang, C.; Allen, R.; Bao, Z. *Nat. Nanotechnol.* **2012**, *7*, 825–832.
- (4) Shepherd, R. F.; Ilievski, F.; Choi, W.; Morin, S. A.; Stokes, A. A.; Mazzeo, A. D.; Chen, X.; Wang, M.; Whitesides, G. M. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108*, 20400–20403.
- (5) (a) Kim, D.-H.; Lu, N.; Ma, R.; Kim, Y.-S.; Kim, R.-H.; Wang, S.; Wu, J.; Won, S. M.; Tao, H.; Islam, A.; Yu, K. J.; Kim, T.-i.; Chowdhury, R.; Ying, M.; Xu, L.; Li, M.; Chung, H.-J.; Keum, H.; McCormick, M.; Liu, P.; Zhang, Y.-W.; Omenetto, F. G.; Huang, Y.; Coleman, T.; Rogers, J. A. *Science* **2011**, *333*, 838–843. (b) White, S. R.; Sottos, N. R.; Geubelle, P. H.; Moore, J. S.; Kessler, M. R.; Sriram, S. R.; Brown, E. N.; Viswanathan, S. *Nature* **2001**, *409*, 794–797. (c) Toohey, K. S.; Sottos, N. R.; Lewis, J. A.; Moore, J. S.; White, S. R. *Nat. Mater.* **2007**, *6*, 581–585. (d) Kim, J.-S.; Lee, Y.-H.; Lee, I.; Kim, T.-S.; Ryou, M.-H.; Choi, J. W. *J. Mater. Chem. A* **2014**, *2*, 10862–10868.
- (6) (a) Blaiszik, B. J.; Kramer, S. L. B.; Grady, M. E.; McIlroy, D. A.; Moore, J. S.; Sottos, N. R.; White, S. R. *Adv. Mater.* **2012**, *24*, 398–

401. (b) Odom, S. A.; Chayanupatkul, S.; Blaiszik, B. J.; Zhao, O.; Jackson, A. C.; Braun, P. V.; Sottos, N. R.; White, S. R.; Moore, J. S. *Adv. Mater.* **2012**, *24*, 2578–2581. (c) Odom, S. A.; Tyler, T. P.; Caruso, M. M.; Ritchey, J. A.; Schulmerich, M. V.; Robinson, S. J.; Bhargava, R.; Sottos, N. R.; White, S. R.; Hersam, M. C.; Moore, J. S. *Appl. Phys. Lett.* **2012**, *101*, 043106.
- (7) Hur, J.; Im, K.; Kim, S. W.; Kim, J.; Chung, D.-Y.; Kim, T.-H.; Jo, K. H.; Hahn, J. H.; Bao, Z.; Hwang, S.; Park, N. *ACS Nano* **2014**, *8*, 10066–10076.
- (8) Wojtecki, R. J.; Meador, M. A.; Rowan, S. J. *Nat. Mater.* **2011**, *10*, 14–27.
- (9) (a) Chakrabarty, R.; Mukherjee, P. S.; Stang, P. J. *Chem. Rev.* **2011**, *111*, 6810–6918. (b) Leininger, S.; Olenyuk, B.; Stang, P. J. *Chem. Rev.* **2000**, *100*, 853–908.
- (10) (a) Wu, N.-W.; Chen, L.-J.; Wang, C.; Ren, Y.-Y.; Li, X.; Xu, L.; Yang, H.-B. *Chem. Commun.* **2014**, *50*, 4231–4233. (b) Li, Z.-Y.; Zhang, Y.; Zhang, C.-W.; Chen, L.-J.; Wang, C.; Tan, H.; Yu, Y.; Li, X.; Yang, H.-B. *J. Am. Chem. Soc.* **2014**, *136*, 8577–8589. (c) Cordier, P.; Tournilhac, F.; Soulie-Ziakovic, C.; Leibler, L. *Nature* **2008**, *451*, 977–980.
- (11) (a) Harano, K.; Hiraoka, S.; Shionoya, M. *J. Am. Chem. Soc.* **2007**, *129*, 5300–5301. (b) Freye, S.; Michel, R.; Stalke, D.; Pawliczek, M.; Frauendorf, H.; Clever, G. H. *J. Am. Chem. Soc.* **2013**, *135*, 8476–8479. (c) Freye, S.; Hey, J.; Torras-Galán, A.; Stalke, D.; Herbst-Irmer, R.; John, M.; Clever, G. H. *Angew. Chem., Int. Ed.* **2012**, *51*, 2191–2194.
- (12) Pan, L.; Yu, G.; Zhai, D.; Lee, H. R.; Zhao, W.; Liu, N.; Wang, H.; Tee, B. C.-K.; Shi, Y.; Cui, Y.; Bao, Z. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109*, 9287–9292.
- (13) Shi, Y.; Ma, C.; Peng, L.; Yu, G. *Adv. Funct. Mater.* **2015**, *25*, 1219–1225.
- (14) Wang, M.; Wang, C.; Hao, X.-Q.; Li, X.; Vaughn, T. J.; Zhang, Y.-Y.; Yu, Y.; Li, Z.-Y.; Song, M.-P.; Yang, H.-B.; Li, X. *J. Am. Chem. Soc.* **2014**, *136*, 10499–10507.
- (15) Lai, Y.-T.; Reading, E.; Hura, G. L.; Tsai, K.-L.; Laganowsky, A.; Asturias, F. J.; Tainer, J. A.; Robinson, C. V.; Yeates, T. O. *Nat. Chem.* **2014**, *6*, 1065–1071.
- (16) (a) Shi, Y.; Pan, L.; Liu, B.; Wang, Y.; Cui, Y.; Bao, Z.; Yu, G. *J. Mater. Chem. A* **2014**, *2*, 6086–6091. (b) Pan, L.; Chortos, A.; Yu, G.; Wang, Y.; Isaacson, S.; Allen, R.; Shi, Y.; Dauskardt, R.; Bao, Z. *Nat. Commun.* **2014**, *5*, 3002.
- (17) (a) Du, M.; Zhang, Z.-H.; Zhao, X.-J.; Xu, Q. *Inorg. Chem.* **2006**, *45*, 5785–5792. (b) Melikova, S. M.; Rutkowski, K. S.; Gurinov, A. A.; Denisov, G. S.; Rospenk, M.; Shenderovich, I. G. *J. Mol. Struct.* **2012**, *1018*, 39–44.
- (18) Joseph, J.; Jemmis, E. D. *J. Am. Chem. Soc.* **2007**, *129*, 4620–4632.
- (19) Zhao, X. *Soft Matter* **2014**, *10*, 672–687.
- (20) Chakraborty, P.; Bairi, P.; Roy, B.; Nandi, A. K. *ACS Appl. Mater. Interfaces* **2014**, *6*, 3615–3622.
- (21) Qiu, L.; Liu, D.; Wang, Y.; Cheng, C.; Zhou, K.; Ding, J.; Truong, V.-T.; Li, D. *Adv. Mater.* **2014**, *26*, 3333–3337.
- (22) Taylor, M. J.; Tanna, S.; Sahota, T. S.; Voermans, B. *Eur. J. Pharm. Biopharm.* **2006**, *62*, 94–100.
- (23) Wu, H.; Yu, G.; Pan, L.; Liu, N.; McDowell, M. T.; Bao, Z.; Cui, Y. *Nat. Commun.* **2013**, *4*, 1943.
- (24) Shi, Y.; Peng, L. L.; Ding, Y.; Zhao, Y.; Yu, G. H. *Chem. Soc. Rev.* **2015**, DOI: 10.1039/C5CS00362H.